

SIERRA BLANCA, NM, USA

WMO: 722683

Lat: 33.467N

Lon: 105.533W

Elev: 2076

StdP: 78.75

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 93083

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-9.8	-7.4	-22.6	0.6	3.8	-20.1	0.8	4.6	18.2	6.7	15.5	7.6	3.0	140	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	13.1	31.0	13.3	29.1	13.2	27.9	13.4	17.9	22.3	17.2	22.1	16.5	22.0	5.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	15.8	19.3	16.0	14.7	18.6	14.9	13.7	18.1	60.1	22.0	56.9	21.9	54.8	22.8	20.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	11.8	10.1	-13.6	33.4	3.5	1.0	-16.1	34.2	-18.2	34.8	-20.1	35.4	-22.6	36.1		
(5)			-14.3	18.7	3.0	1.1	-16.5	19.6	-18.2	20.2	-19.9	20.8	-22.0	21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	3.7	4.9	8.1	11.5	16.2	20.8	21.3	20.1	17.6	12.6	7.6	3.4
	DBStd	7.62	4.60	5.10	4.52	4.40	4.09	2.83	2.23	2.20	2.91	4.22	4.75	4.71
	HDD10.0	807	197	151	89	36	6	0	0	0	1	23	99	206
	HDD18.3	2468	453	375	317	206	87	9	3	8	46	179	323	462
	CDD10.0	1672	3	9	31	81	198	325	350	313	229	105	26	2
	CDD18.3	293	0	0	0	1	22	84	96	63	24	2	0	0
	CDH23.3	2642	0	0	1	13	261	924	768	443	207	24	0	0
	CDH26.7	680	0	0	0	1	52	302	211	91	22	1	0	0
(14) Wind	WSAvg	4.0	4.1	4.8	4.8	5.4	4.8	4.0	3.0	2.7	3.0	3.7	3.8	4.2
(15) Precipitation	PrecAvg	417	19	18	19	19	28	31	88	73	53	31	14	24
	PrecMax	559	44	59	47	75	102	102	183	188	107	102	43	79
	PrecMin	212	1	1	3	3	2	2	29	19	9	0	0	1
	PrecStd	79	13	13	10	18	28	27	35	35	26	24	13	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.3	19.0	22.4	25.0	30.2	32.8	32.4	31.0	28.0	26.1	21.4	17.3
		MCWB	5.8	6.9	7.8	9.3	12.4	13.5	14.2	14.3	13.7	11.5	8.3	5.4
	2%	DB	14.1	17.4	20.0	22.8	27.9	31.3	30.2	28.8	27.4	23.7	19.1	14.1
		MCWB	4.3	5.7	6.9	8.2	11.1	12.7	14.2	14.4	13.5	10.9	7.4	4.5
	5%	DB	12.6	15.1	18.0	22.0	26.3	29.9	28.9	27.5	26.0	22.3	17.5	12.6
		MCWB	3.7	4.6	5.9	8.0	10.2	12.4	14.4	15.0	13.1	10.4	6.7	3.6
	10%	DB	11.0	12.9	17.0	19.9	24.1	28.0	27.6	26.1	24.0	20.0	15.1	10.9
		MCWB	3.0	3.5	5.5	7.0	9.4	12.1	14.7	15.1	13.1	9.7	5.6	2.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	8.1	9.7	12.4	15.2	17.3	18.8	19.1	17.8	15.7	10.4	7.4
		MCDB	13.8	15.8	18.3	18.5	20.3	24.4	23.0	22.9	21.6	19.0	16.9	12.9
	2%	WB	5.8	6.7	8.0	10.2	13.5	16.1	17.8	18.1	16.6	13.9	8.7	5.6
		MCDB	12.1	14.7	16.7	18.8	21.7	22.7	22.7	22.0	20.9	18.9	15.7	12.6
	5%	WB	4.5	5.5	6.9	9.0	12.4	15.3	17.0	17.3	15.8	12.5	7.7	4.4
		MCDB	11.2	13.6	16.0	18.0	21.5	23.0	22.6	21.7	20.6	18.2	14.8	11.2
	10%	WB	3.4	4.3	5.7	7.9	11.3	14.5	16.4	16.6	15.0	11.2	6.6	3.3
		MCDB	9.7	12.2	15.0	17.6	20.7	22.6	22.6	21.7	20.3	17.5	14.2	9.6
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.5	12.6	12.9	12.7	13.1	11.8	11.1	11.4	11.8	11.6	10.7
		MCDBR	12.4	13.3	14.2	14.3	14.4	14.7	13.7	12.7	13.1	13.3	13.2	12.2
	5% WB	MCWBR	6.8	6.8	7.0	6.8	6.4	6.0	4.8	4.6	5.0	6.2	6.8	6.6
		MCDWR	11.5	12.2	13.4	13.7	13.0	12.6	11.2	10.3	10.6	11.1	12.0	11.3
(40) Clear-Sky Solar Irradiance	taub		0.227	0.233	0.250	0.267	0.284	0.309	0.341	0.324	0.294	0.263	0.242	0.231
		taud	2.704	2.690	2.627	2.565	2.569	2.509	2.475	2.503	2.569	2.657	2.655	2.682
	Ebn at Noon		1026	1044	1039	1023	999	969	936	951	974	992	994	1002
		Edh at Noon	68	77	89	100	101	107	110	105	94	79	71	66
(44) All-Sky Solar Radiation	RadAvg		3.51	4.50	5.77	7.08	7.61	7.47	6.62	6.12	5.62	4.69	3.72	3.12
	RadStd		0.26	0.38	0.34	0.38	0.39	0.53	0.42	0.35	0.32	0.43	0.27	0.19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	-1.94	N/A	-0.68	-1.20	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page